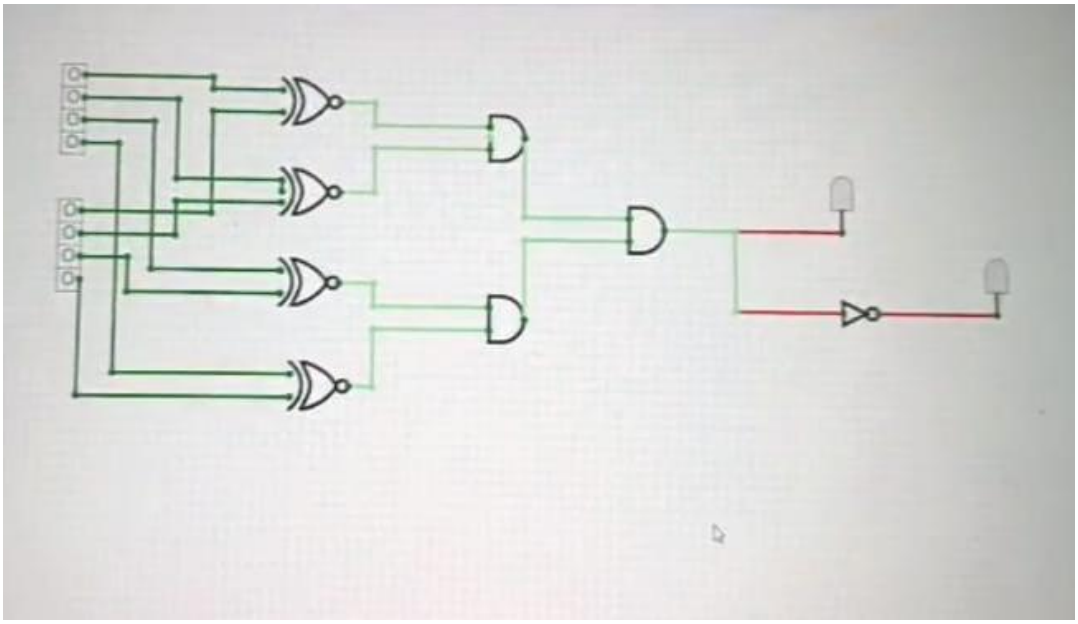
Password project

Circuit design :



K maps :

A	B	output
0	0	1
0	1	0
1	0	0
1	1	1

A’ B’ + AB (XNOR gate)

A	B	output
0	0	0
0	1	0
1	0	0
1	1	1

AB (AND gate)

Component list:

component	function
1 XNOR gate	It gives 1 when two input are same
1 AND gate	It gives 1 when two input are 1
1 NOT gate	When circuit give 0 not gate covert it to 1
Jumper wires	To connect all component to each other
2 DIP switches 4 way	To control the input (0 or 1) by user
1 green led	It lights up when inputs are same
1 red led	It lights up when inputs aren't same
resistor	To control voltage input
battrery	Volt source
Breadboard	The main connecter all gates to each other

Cost of circuit:

- [ ] IC 4077 - Quad Exclusive NOR XNOR Gate x1 (20.00 EGP)

- [ ] IC 7408 - Quad 2 input AND Gates - Original x1 (15.00 EGP)

- [ ] IC 7404 - NOT Logic Inverter - Original x1 (15.00 EGP)

- [ ] BB-01 Breadboard 830 Tie Point x2 (70.00 EGP)

- [ ] DIP Switch 4 Way x2 (20.00 EGP)

- [ ] Kit Resistors 1/4W -100 Carbon Resistors 1 % Tolerance 1/4W x1 (25.00 EGP)

- [ ] PH61 -20cm Male to Male 40 Jumper Wires Set x1 (35.00 EGP)

- [ ] LED 5mm Red Color x1 (0.50 EGP)

- [ ] LED 5mm Green Color x1 (0.50 EGP)

Total: 201.00 EGP